MODULE-IV

TOPICS

- DYNAMIC PROGRAMMING-INTRODUCTION
- CONTROL ABSTRACTION-DYNAMIC PROGRAMMING
- DAndC VS DP

PROBLEMS

1. ALL-PAIRS SHORTEST PATH PROBLEM
2. TRAVELLING SALES PERSON PROBLEM

DYNAMIC PROGRAMMING

- Dynamic Programming is used in **optimization problems**.
- Like divide-and-conquer method, Dynamic Programming solves problems by **combining the solutions of sub problems**.
- Dynamic Programming algorithm **solves each sub-problem just once** and then **saves its answer in a table**. Thereby avoiding the work of re-computing the answer every time.
- Two main properties of a problem suggest that the given problem can be solved using Dynamic Programming.

I. **Overlapping sub-problems**
II. **Optimal substructure.**

Overlapping Sub-Problems

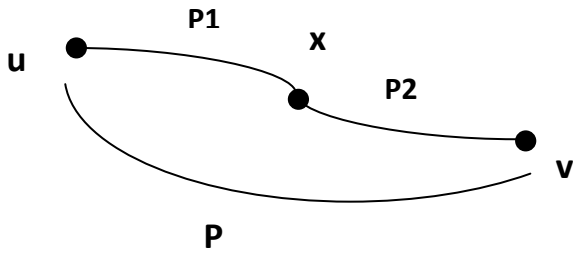
- Dynamic Programming combines solutions to sub-problems.
- It is mainly used where the solution of one sub-problem is needed repeatedly.

- The computed solutions are stored in a table, so that these don't have to be re-computed. Hence, this technique is needed where overlapping sub-problem exists.
- For example, recursive program of Fibonacci numbers have many overlapping sub-problems.

Optimal Sub-Structure

- A given problem has Optimal Substructure Property, if the optimal solution of the given problem can be obtained using optimal solutions of its sub-problems.
- For example, the Shortest Path problem has the following optimal substructure property –

✓ If a node x lies in the shortest path from a source node u to destination node v, then the shortest path from u to v is the combination of the shortest path from u to x, and the shortest path from x to v.



P1: Shortest path from u to x

P2: Shortest path from x to v

P: Shortest Path from u to v (Combination of P1 & P2)

Steps of Dynamic Programming Approach

Dynamic Programming algorithm is designed using the following four steps −

1. Characterize the structure of an optimal solution.
2. Recursively define the value of an optimal solution.
3. Compute the value of an optimal solution, typically in a bottom-up fashion.
4. Construct an optimal solution from the computed information.

Problems

- **All-pairs Shortest Path Problem**
- **Travelling Salesperson Problem**

CONTROL ABSTRACTION FOR DYNAMIC PROGRAMMING

DP (n)

{

Optimal solution OS[n] //data structure for optimal solution

for i ← 1 to n

OS[i] ← compute (OS [1], OS [2] … OS [j]), j<i

return OS[n]

}

DIVIDE AND CONQUER VS DYNAMIC PROGRAMMING

Divide& Conquer	Dynamic Programming
1. It is Recursive.	1. It is non Recursive.
2. It does more work on sub problems and hence has more time consumption.	2. It solves sub problems only once and then stores in the table.
3. It is a top-down approach.	3. It is a Bottom-up approach.
4. In this sub problems are independent of each other.	5. In this sub problems are interdependent.
6. For example: Merge Sort & Binary Search etc.	6. For example: Travelling Salesman